Hospital Quality Analysis

CMS Hospital Compare — 5,400+ U.S. Hospitals

Three questions. A bronze-silver-gold pipeline. Surprising answers.

Built with **PostgreSQL 14 · SQL · Tableau** | Portfolio Project by **Vladimir Mickic**

EXECUTIVE SUMMARY

Three Hypotheses Tested. All Three Were Wrong.

Every initial assumption was overturned by the data — revealing that hospital quality problems are more fragmented, and more nuanced, than conventional wisdom suggests.

r = −0.020

Imaging overuse and ED wait times are **statistically independent**. R^2 = 0.0004 across 3,589 hospitals. They are entirely separate problems.

221 min

Non-Profits have the **highest star ratings** (3.18 avg) but the **slowest EDs** (221 min). Clinical quality ≠ operational speed.

83% → 5 "Worse"

A hospital scored 83% on process compliance and still earned 5 **"Worse"** complication ratings. Checklists ≠ safety.

Imaging overuse and ED throughput are separate problems. They need separate interventions and separate budgets.

THE PROBLEM

Why This Analysis Exists

CMS publishes quality data on 5,400+ hospitals — but it arrives in **5 separate files** with inconsistent formats, suppressed scores, and mixed data types in the same columns. The raw data is rich but unusable without a structured pipeline.

This analysis builds that pipeline and uses it to answer three high-stakes questions about how hospital quality actually works.

1

Imaging vs. ED Waits

Do hospitals that over-order imaging also have longer ED wait times?

2

Ownership Type

Does ownership structure predict quality patterns? Do incentive structures matter?

3

Compliance vs. Safety

Can a hospital follow every protocol and still be unsafe for patients?

METHODOLOGY

Bronze → Silver → Gold Pipeline

A three-tier architecture transforms 430K+ raw rows from 5 CMS CSV files into clean, analytical views that directly answer each business question.

1

🥉 Bronze — Raw Landing

All columns stored as TEXT. No casting, no transforms. 5 tables, 430K+ rows. Per-table exception handling with row-count logging.

2

🥈 Silver — Cleaned & Typed

Type casting, NULL standardization, tiered score usability flags. 12 ownership values collapsed to 4 canonical groups. Multi-code footnote parsing via array overlap.

3

🥇 Gold — Analytical

3 focused views answering the 3 questions. 1 hospital dimension view with Census regions. No monolithic fact table — each view joins only what it needs.

Table	Rows	Domain
cms_hospital_general	5,426	Ratings, ownership, location
cms_timely_care	138,129	ED waits, vaccination, timing
cms_complications	95,780	Mortality & complications
cms_outpatient_imaging	18,500	CT/MRI ordering efficiency
cms_infections	172,404	HAI rates

FINDING 1

No Link Between Imaging Overuse & ED Wait Times

−0.020

Pearson r

Near-zero correlation across 3,589 hospitals

0.0004

R^2 Value

Imaging explains 0.04% of ED wait time variance

What Was Tested

- **Imaging composite:** AVG of OP-10, OP-13, OP-39 (lower % = better). OP-8 excluded — its mean (36.9%) is 5× higher than peers and would dominate the composite.
- **ED wait:** AVG of OP_18a–d in minutes, capped at 1,440
- **Sample:** n = 3,589 hospitals with usable scores in both domains

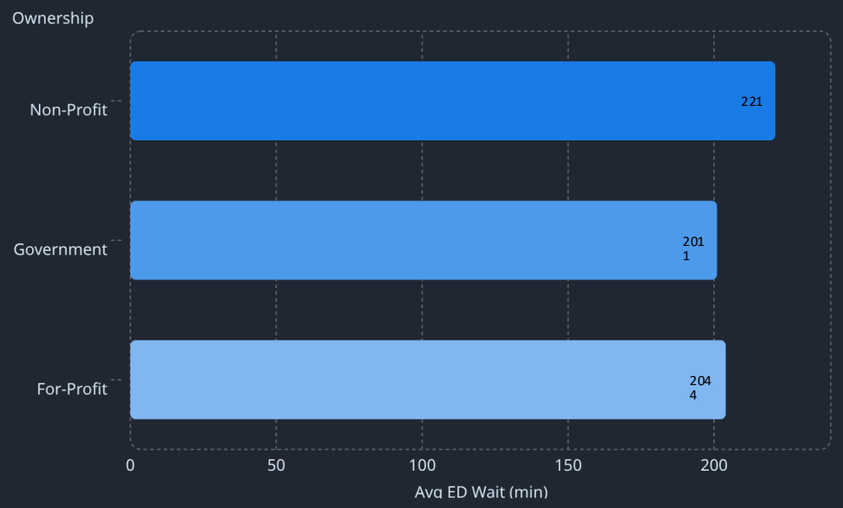
Insight: Imaging overuse is a clinical ordering decision. ED throughput is an operational flow problem. Different root causes → different interventions required.

FINDING 2

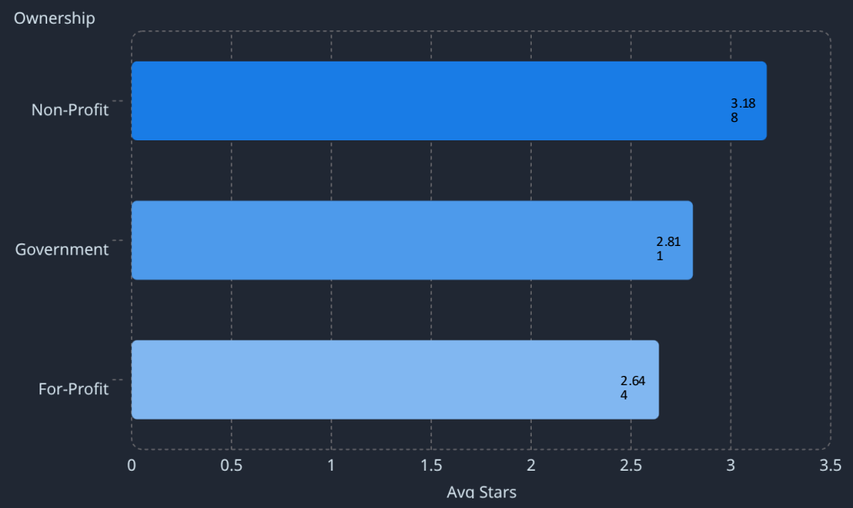
The Non-Profit Paradox

Non-Profits lead on clinical star ratings — but run the slowest emergency departments of any ownership group. The correlation between imaging and ED waits stays near zero in every group.

Avg ED Wait Time (min)



Avg Star Rating (out of 5)



The Core Paradox

Non-Profits optimize for clinical quality metrics — mortality, readmissions, patient experience. This may come at the cost of operational throughput. Star ratings and ED speed measure fundamentally different things.

Methodology Caveat

Non-Profit share rises from 54% → 64% in the joined sample due to higher reporting rates — a form of survivorship bias. Tribal hospitals (collapsed to 1 entity) were excluded from group analysis.

FINDING 3

Compliance Does Not Equal Safety

Among the 30 worst-complication hospitals (3+ "Worse" ratings), compliance scores range from 42%–84% with no clear pattern.

Correlation Test	r value	Relationship
Process compliance vs. Complications	≈ 0	None
Process compliance vs. Mortality	≈ 0	None
Process compliance vs. Patient Safety (PSI)	0.08	Barely detectable

Hospital 1800445 "Worse" complication ratings
Process compliance: **83.55%** — High compliance, still deeply unsafe.

Hospital 0501454 "Worse" complication ratings
Process compliance: **42.17%** — Low compliance AND unsafe, a different failure mode entirely.

Process measures track narrow boxes — colonoscopy care, staff vaccination. Clinical outcomes depend on staffing, communication, equipment, and patient acuity. **Checklists don't capture these factors.**

STRATEGIC RECOMMENDATIONS

Two Interventions. Separate Budgets. Measurable Targets.

The data points to two structurally independent problems — each requiring its own roadmap, ownership, and success metrics.

Imaging Utilization Reform

Target OP-10 ordering rates. Implement radiologist-led pre-authorization protocols for contrast CT. Benchmark against the 4–5% rate seen in 5-star hospitals. Expected impact: 30–50% reduction in unnecessary contrast orders within 12 months.

CLINICAL ORDERING · LOW COST · HIGH SIGNAL

ED Throughput Optimization

ED wait times are operationally driven — staffing models, triage protocols, and bed management. Treat as a separate workstream from imaging. Expected impact: 15–25 min reduction in median wait time with lean process redesign.

OPERATIONAL · SEPARATE BUDGET · QUICK WIN

These are not the same problem. Conflating them wastes resources. Each intervention has a distinct owner, timeline, and KPI.

DEEP DIVE

Ownership Is Not a Monolith

Ownership Subtype	Avg Stars	Notable Pattern
VA Hospitals	**4.2**	77% are high performers
Non-Profit (all subtypes)	~3.2	~40% high performers
Physician-Owned (For-Profit)	3.32	Outperforms its sector
Local/State Government	2.7–2.8	Disproportionate low performers
Proprietary (For-Profit)	2.55	~49% low performers

The 1.5-Star Gap

Lumping VA hospitals with local government hospitals under "Government" masks a **1.5-star quality gap** — one of the largest differences in the entire dataset.

Reporting ≠ Quality

Reporting completeness vs. star ratings: **r = −0.041** (R^2 = 0.0017). Hospitals that report on more measures do not score better. Transparency and quality are independent dimensions.

CASE STUDY

UPMC Presbyterian Shadyside — Gap to 5 Stars

Benchmarked against Pennsylvania non-profit acute care peers (facility 390164, current 4-star rating). UPMC excels in high-acuity ED pathways — **133–149 min faster** than PA average on OP_18c/d — but two specific gaps prevent a 5-star rating.

Measure	UPMC	5-Star Avg	Gap
PSI_03 — Pressure Ulcers	1.84	0.50	**+1.34 — nearly 4× worse**
OP-10 — CT Contrast Overuse	12.2%	~4–5%	**Nearly double PA rate**
HAI_1_SIR — Central Line Infections	Above benchmark	Below benchmark	Gap exists
PSI_90 — Safety Composite	Elevated	Lower	Driven by PSI_03

No UPMC hospital achieved 5 stars in PA. The 5-star group is dominated by smaller regional systems: **WellSpan, St. Luke's, AHN Wexford**. Two actionable problems — pressure ulcer protocols and CT contrast ordering are specific, measurable, and addressable.

PIPELINE QUALITY & WRAP-UP

Bugs Fixed, Limitations Acknowledged

Bug	Impact	Fix Applied
Footnote matching used exact strings	Missed multi-code values like '5, 23'	Switched to array overlap (&&)
Score exclusion priority reversed	Footnoted rows misclassified	Footnotes evaluated before NULL scores
Zero-infection hospitals not flagged	Code 8 rows treated as valid scores	Added to usability blacklist
birthing_friendly conflated unknown/no	Unknown hospitals marked as "No"	NULL for unknowns

Data Quality Realities

- Rural suppression: MT, SD, WY → 65–80% unusable vs. 15–30% for NJ, CT, RI
- 58% of timely_care scores = "Not Available"
- 47% of hospitals have no star rating
- Only 11.9% have usable scores across all 6 HAI measures